

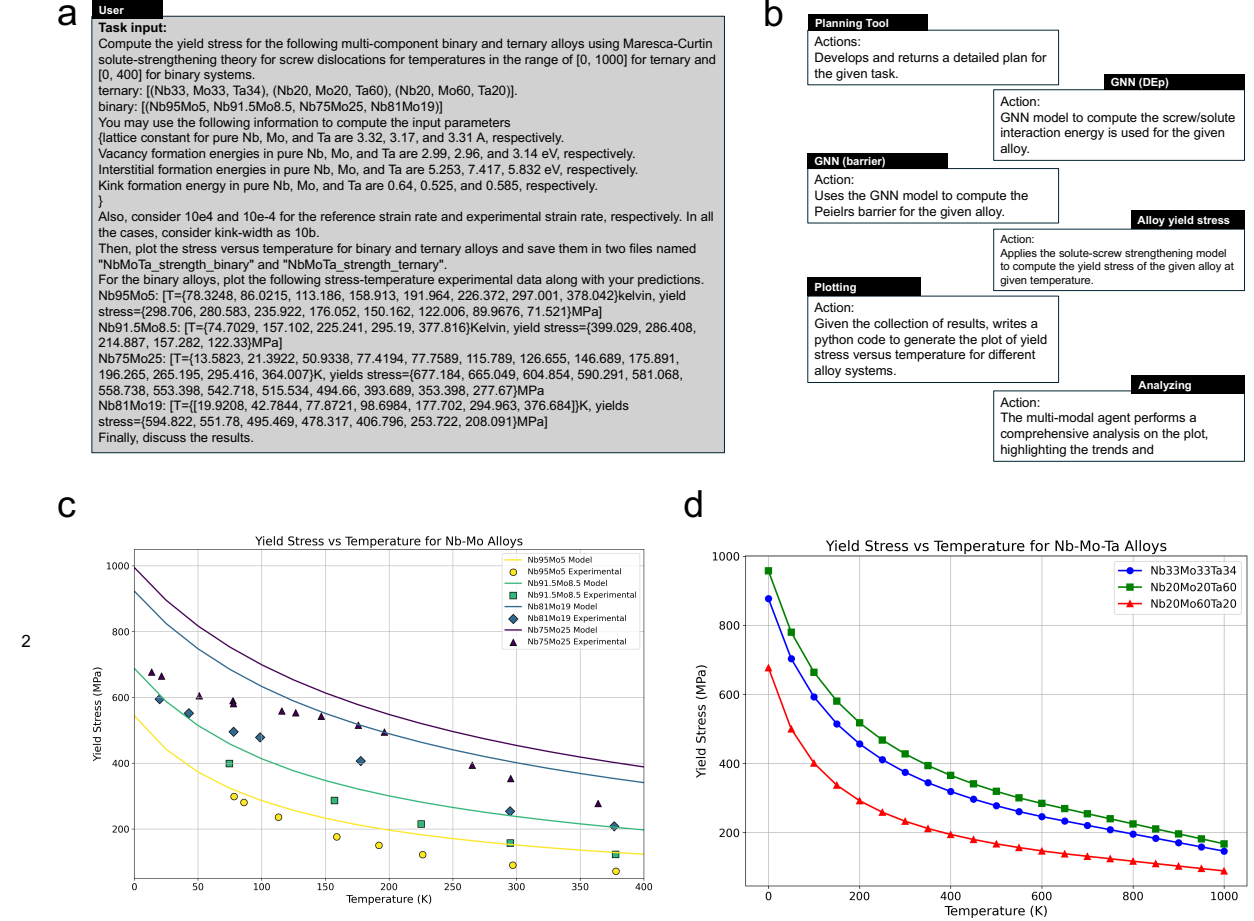


Figure 7: Overview of the multi-agent collaboration to predict the yield stress in binary and ternary alloy BCC systems. (a) The input task involving computing the yield stress for a set of compositions. Some material properties for the pure metals are provided as well as the strain rates. Moreover, the experimental data for binary alloys are provided to compare with the predictions. (b) The workflow of computations performed by our multi-agent system featuring key tasks like planning, tool calling for material property predictions from the GNN model, and finally plotting and analyzing the results. (c) Predictions of our model for the ternary alloys and (d) the predictions of our model for binary alloys along with the experimental results.

for other input parameters such as the lattice constant, kink formation energy, and vacancy/interstitial formation energies, the system computes their values by averaging over the pure elements. Once all necessary input parameters are obtained, they are fed into the yield stress prediction tool for final calculations. The results are then passed to the coding tool, which generates Python code to plot the predictions. Finally, the multi-modal agent, powered by the GPT-4o LLM, analyzes the resulting plots to evaluate trends across compositions.

Despite the complexity of the task, which requires numerous calculations and input parameters, we observe that all sub-tasks are executed seamlessly. The primary outcomes of the multi-agent collaboration are presented in Figures 7(c) and (d), which show the variation of yield stress with temperature for binary and ternary systems, respectively. The binary plot also includes experimental data from the input task for comparison.

Following this, the multi-modal agent conducts a comprehensive analysis of the results, as detailed in Figures S6 and S7 in the Supplementary Information (SI), covering binary and ternary alloys, respectively. Upon completing the analysis, the assistant provides a summary of key insights, which is presented in Figure 8.